INSTRUCTIONS

# AF—400*

# DYNAMIC SLOWDOWN MODULE

*These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.*

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## DESCRIPTION

The Dynamic Slowdown Module is an option for use with the AF-400* Inverter Drive in order to provide additional braking torque for decelerating the AC motor and connected load.

The module is basically a static switch used to connect a DB (dynamic braking) resistor across the DC link as shown in Figure 1.

**NOTE:** Refer to the AF-400* Inverter Drive Instruction Book for information regarding the basic Inverter operation.

## START— UP

It is recommended that initial checkout of the drive is first performed with the DB module disconnected such that proper inverter operation can be verified. This can be done by disconnecting and taping up the power lead from the resistor DBR at the terminal BRA on the braking module and unplugging the flat cable harness connector CN1 from the slowdown control PC card.

After proper inverter operation is established, with power to the drive disconnected, reconnect the power wire to BRA and reconnect the CN1 connector.

Now verify that all connections are made as shown on the elementary diagram as follows:

BRA Connects to the DB resistor with the other side of the resistor connected to the positive DC link.

P2A Connects to the negative DC link.

CN1 Check that the flat cable connector is properly inserted.

CN2 Connects to the CS SCR with the white gate lead at CG (WHT).

CN3 Connects to the DBS SCR with the white gate lead at DG (WHT).

CN4 Check that the red lead connects to the RP point.

**Note that the CN1-4 connectors should be inserted such that the locking feature is properly engaged.**

TB1 If used, support this terminal board to prevent braking any solder joints when making connection. (The pilot duty rating of the RX relay contacts is .3A @ 115V AC)

TB2 The internal wiring connects through 3M ohm resistors in each lead to the CN4 connector. The red wire connects to RP which again connects to the negative side (same as BRA) of the DB resistor.

W1-W4 Check that the wires from the SCR heatsinks are properly connected to the stab-on posts on the RC snubber card.

Reapply power and run the motor up to about 50% speed. Quickly reduce the reference to zero. If the load inertia is low it may be necessary to reduce the deceleration time in order to activate the DB operation.

## OPERATION

Figure 2 shows a detailed diagram of the power circuit. Turning on the SCR DBS closes the "switch" to connect the resistor R across the DC link. The subsequent firing of SCR CS commutates the DBS SCR off to open the "switch".

This on-off chopper control permits regulation of the amount of power dissipated in the DB resistor and consequently the amount of braking torque applied to the motor.

When the AC motor is given a slowdown command by decreasing the applied frequency the rotating inertial energy is partially absorbed by friction and windage losses, partially by electrical losses ($I^2R$) in the motor windings and the inverter commutation circuitry. These losses will typically produce a braking torque from 10% to 20% of rated motor torque. If the rate of deceleration requires more braking torque the motor will regenerate energy back into the DC link and thereby increase the voltage on the link filter capacitor C1.

In the case where no means to dissipate the energy is provided the inverter control will detect the increase in C1 voltage and the rate of deceleration is slowed down (Refer to the main Instruction book). The function of the Dynamic Slowdown Module is to connect the DB resistor across the DC link to discharge its excess voltage by an appropriate chopping action such that the volts per Hertz (V/Hz) ratio of the motor is maintained at a correct level during deceleration.

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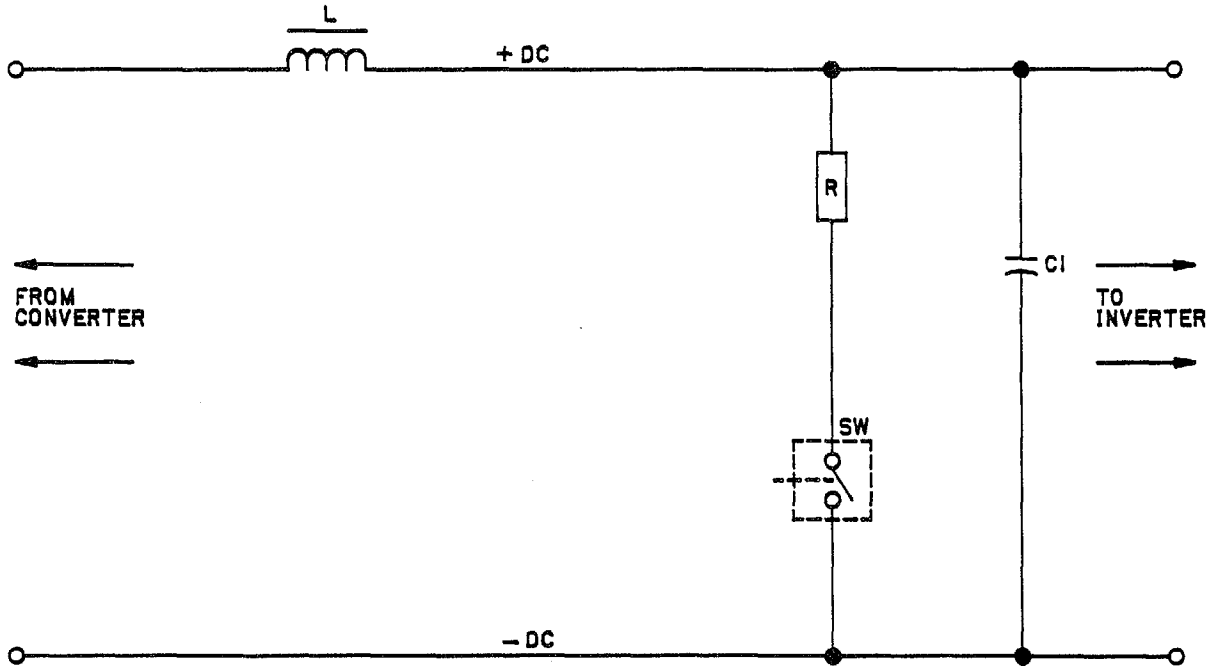



**FIGURE 1**
SIMPLIFIED CIRCUIT

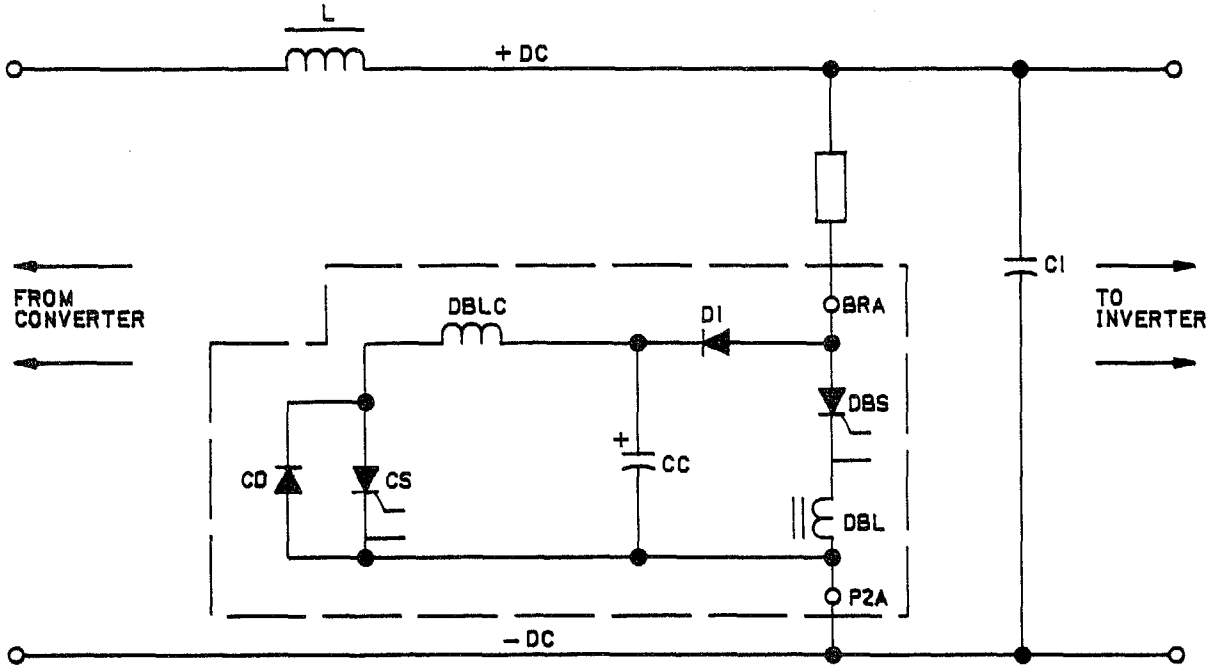



**FIGURE 2**
DYNAMIC SLOWDOWN POWER CIRCUIT

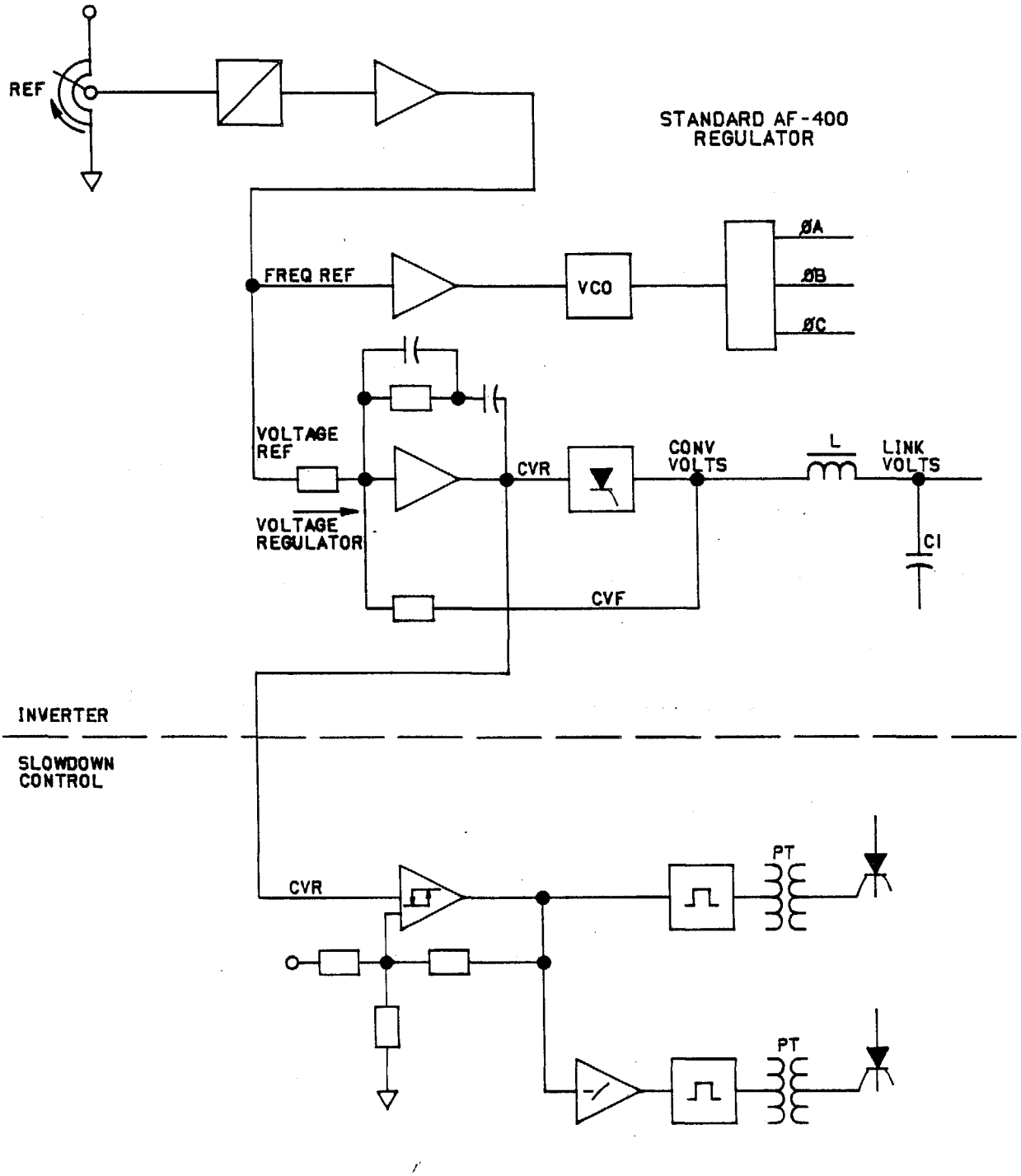



**FIGURE 3**
CONTROL CIRCUIT BLOCK DIAGRAM

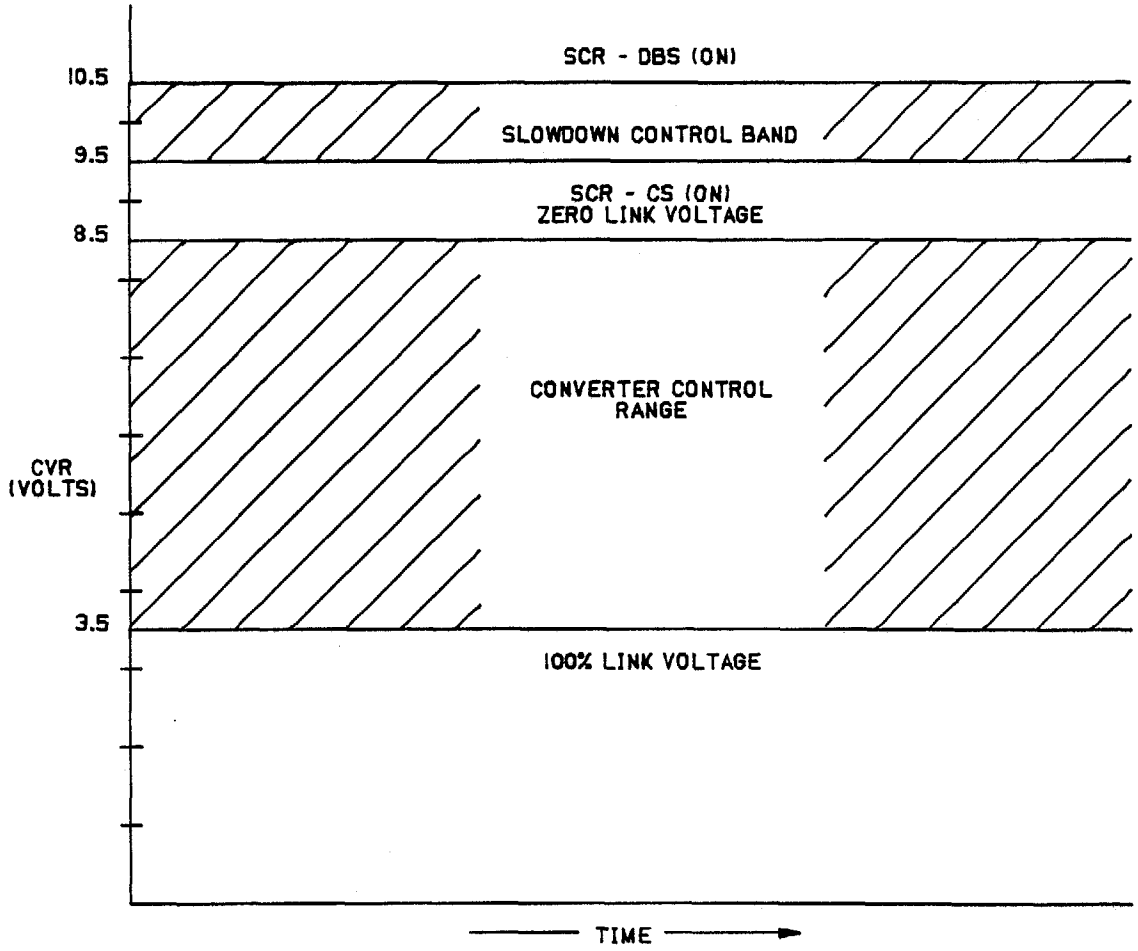



**FIGURE 4**
SLOWDOWN MODULE/CONVERTER CONTROL RANGE

## CONTROL

The control for the Slowdown Module is shown in principle on Figure 3 and Figure 4.

The Inverter voltage regulator error signal, CVR, which normally controls the phase advance (firing angle) of the converter is also used to control the firing of the dynamic braking SCR, DBS, and the commutation SCR, CS.

During acceleration of the motor from minimum to rated voltage and frequency the CVR voltage will decrease from about 8.5 volts to about 3.5 volts. Now, if the motor is decelerated at a rate fast enough to increase the voltage on the DC link to a level above what the voltage reference calls for, the converter voltage feedback, CVF, is also increased such that the converter is phased back (turned off). The phase back occurs as the CVR voltage increases back to the 8.5 volt level. During this process the converter and voltage regulator no longer control the link voltage, however, as CVR increases to about 10.5 volts a comparator in the Slowdown Control card will switch and produce a firing signal to the dynamic braking SCR, DBS connecting the DB resistor across the DC link such that the voltage starts decreasing. The converter voltage feedback detects the reduction in DC link voltage and will cause the error voltage to start to decrease.

The initial switching of the comparator reduced it's bias level such that the CVR voltage must be reduced to about 9.5 volts before the comparator output switches back again. As the comparator switches back, however, a firing pulse is applied to the commutation SCR, CS which turns off the DBS SCR disconnecting the DB resistor from the DC link.

The process will now repeat. As the regenerative action of the motor again increases the C1 voltage the CVR voltage starts increasing and results in a refiring of the DBS SCR as CVR again reaches the 10.5 volt level.

This chopping action will continue with a gradual increase in on-time versus off-time of the DB resistor as the frequency and voltage is reduced. At some point, depending on the load friction, inertia and the deceleration rate of the reference, when the frequency and voltage has been reduced to a certain level, the DB resistor will be on continuously and the braking torque starts tapering off proportional to the speed which will result in a tailout to stop.

### THE TURN-OFF PROCESS

Figure 5 shows the wave shapes of currents and voltages during the off-commutation of both the DBS and the CS SCRs. The time period shown covers a time span of about 100 microseconds.

During normal motoring operation the commutation capacitor CC is charged up through the DB resistor and diode D1 to a voltage equal to the DC link voltage across C1.

Prior to time **t0** the DBS SCR is turned on, to connect the DB resistor across the DC link.

At time **t0** the CS SCR is turned on, current (**IC** and **IL**) will build up in the LC circuit through CC, DBLC and CS. As the current builds up the CC voltage **VC** starts dropping.

The DB current **IR & IDBS** through the resistor and the DBS SCR stays essentially constant until time **t1**.

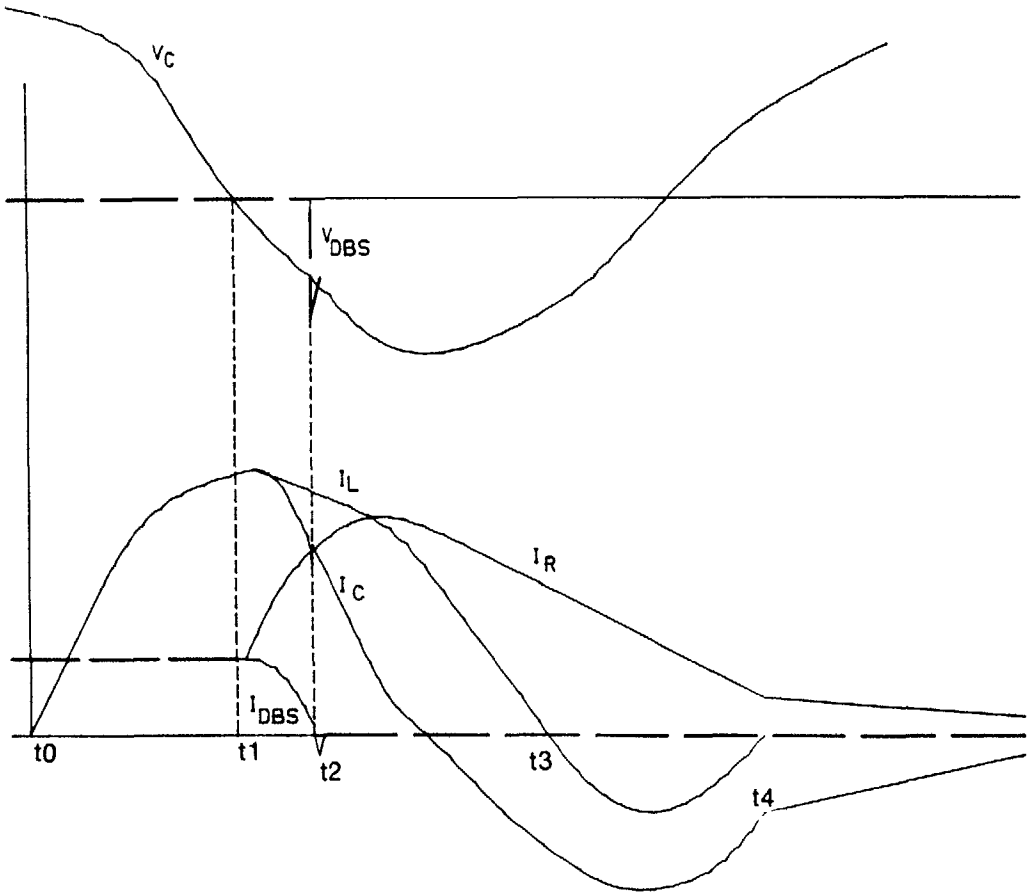
At this time the capacitor voltage, **VC**, reverses making the current **IDBS** start reducing as the diode D1 starts conducting.

At time **t2 IDBS** reaches zero and the DBS SCR turns off. **VDBS** will now switch, with some overshoot, to equal the now negative capacitor voltage **VC**.

Also at time **t1** when **VC** goes negative the DB resistor current will increase and oppose the current in CC such that the net **IC** current drops below the **IL** current.

At time **t3** the **IL** current is reduced to zero, reverses and starts flowing through the CD diode. This back biases the CS SCR until time **t4**, when the **IL** is again zero. The time period **t3** to **t4** must exceed the turn-off time required to permit the CS SCR to recover and enable it to block the forward voltage being applied as the CC capacitor now is recharged to again equal the C1 voltage. With CC recharged the commutation capability is restored and the DBS SCR may again be turned on.

As the DC link voltage decreases during deceleration the amplitude of the current and voltage signals shown in Figure 5 will decrease proportionally. Consequently the commutation capability of the "switch" is reduced proportional to the DC link voltage and it is therefore not possible to provide increased braking torque by reducing the DB resistance at lower speeds.




VOLTAGE AND CURRENT WAVE SHAPES

$I_C$ - COMMUTATION CAPACITOR CC CURRENT

$V_C$ - COMMUTATION CAPACITOR CC VOLTAGE

$I_L$ - COMMUTATION INDUCTOR DBLC CURRENT

$I_R$ - DB RESISTOR CURRENT

$I_{DBS}$ - DBS SCR CURRENT

$V_{DBS}$ - DBS SCR VOLTAGE

**FIGURE 5**

## RATING

The rating of the Dynamic Slowdown Modules are matched to the respective voltage kVA ratings of the Inverters as shown below.

### NOTE

During deceleration the AC current generated by the motor and fed back through the Inverter to the DC link should not exceed 150% of the continuous current rating of the Inverter. Nor should the instantaneous current through the DB resistor exceed the commutation capability of the slowdown module.

| | 460VAC | | | | 230VAC | | | |
|---|---|---|---|---|---|---|---|---|
| Catalog Number 36D877420 | AAG01 | AAG02 | AAG03 | AAG04 | BAG01 | BAG02 | BAG03 | BAG04 |
| Inverter rating kVA | 100 | 200 | 300 | 400 | 25 | 50 | 75 | 100 |
| Nominal Inverter Current, amps AC | 130 | 260 | 390 | 520 | 70 | 140 | 210 | 280 |
| Maximum DB Current @600VDC, amps DC | 200 | 400 | 600 | 800 | | | | |
| Maximum DB Current @300VDC, amps DC | | | | | 100 | 200 | 300 | 400 |
| Minimum DB resistance, ohms | 3.0 | 1.5 | 1.0 | 0.75 | 3.0 | 1.5 | 1.0 | 0.75 |

Typical applications conforming to industrial standards normally requires an initial maximum braking torque capability of 150% of rated torque.
The initial DB resistor current required and the respective resistance can be determined by the formulas.

**IDB** = (Motor HP x 746) x (% Braking Torque/100) x (1/VDC) x (Efficiency)

**RDB = VDC/IDB**

For 150% torque, and an efficiency of .85 at 600VDC

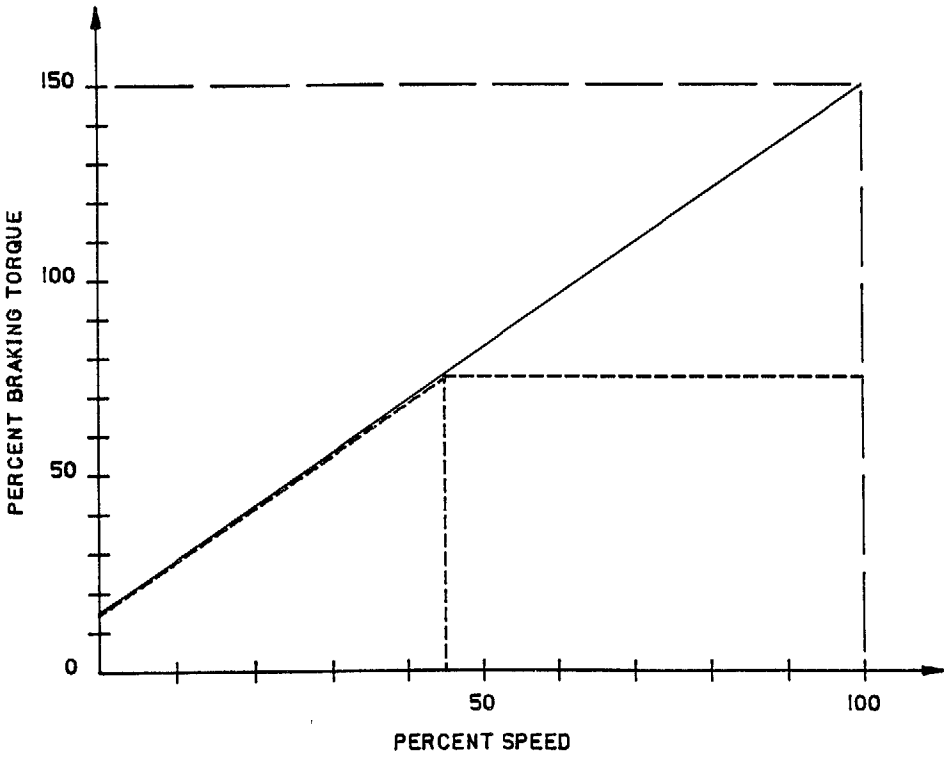
**IDB** = HP x 746 x 1.5 x .85/600 = 1.6 x HP (amps); at 600VDC

**IDB** = HP x 746 x 1.5 x .85/300 = 3.2 x HP (amps); at 300VDC

and

**RDB** = 600/(1.6 x HP) = 379/HP (ohms per HP); at 600VDC

**RDB** = 300/(3.2 x HP) = 95/HP (ohms per HP) at 300VDC




MAX BRAKING TORQUE vs SPEED

(FOR 150% MAX TORQUE AND 15% FIXED LOSSES)

**FIGURE 6**

For each module with the motor HP equal to the inverter kVA:

| | 460 VAC | | | | 230 VAC | | | |
|---|---|---|---|---|---|---|---|---|
| HP (kVA) | 100 | 200 | 300 | 400 | 25 | 50 | 75 | 100 |
| Max. initial DB current@600VDC, amps DC | 160 | 320 | 480 | 640 | | | | |
| Max. initial DB current @300VDC, amps DC | | | | | 80 | 160 | 240 | 320 |
| DB resistance, ohms | 3.8 | 1.9 | 1.3 | 0.95 | 3.8 | 1.9 | 1.3 | 0.95 |

Figure 6 shows graphically how maximum braking torque is reduced with speed under the above conditions of 150% max. torque and with friction and losses assumed fixed at 15%.

As indicated a linear deceleration rate requiring 75% braking torque can be maintained by chopping from 100% to 45% speed. From 45% speed to a stop the DB resistor will be connected across the link all the time and the torque will taper off.

## OVERLOAD PROTECTION

The slowdown control card contains a separate electronic thermal overload circuit to detect if the DB resistor is being subjected to excessive power dissipation which could result in resistor damage

The input circuit is the voltage across the DB resistor fed back by high resistance (3M ohm on 600VDC drives, 1.5M ohm on 300VDC drives) to a differential amplifier. The output DB trip signal DBT connects to the driver point **IFT**. An overload condition makes DBT go low (0 to +1V) to cause a normal drive shutdown.

The indicating light, TOL will turn on and a fault relay, RX which is normally picked up will drop out. A N.O. RX contact is available on the card terminal board points RX1 and RX2.

The fault condition is reset as the reset signal, **RST**, which connects to the drive **RST** signal, goes low.

Since load and motor inertia, motor speed, duty cycle and other factors may vary considerably for a particular application, so will the current/watt rating of the DB resistor used. The trip circuit therefore contains two adjustment potentiometers which, must be properly set depending on the continuous and the short time rating of the DB resistor.

The circuit operates by comparing the DB resistor voltage feedback which also represents resistor current with a preset level, BV, set by BIAS potentiometer. Whenever the resistor feedback voltage exceeds the BV voltage level which is typically set to represent 100% to 115% of the continuous resistor rating an integrating amp starts ramping up its output voltage, IV. Likewise, as the feedback drops below the bias level, the integrator voltage, IV, starts ramping back down again. (In a sense the IV voltage can be considered to represent the temperature of the DB resistor).

When the integrator output, IV, ramps up to a voltage level, TV, preset by the TRIP potentiometer, the output comparator amplifier switches from a high to a low voltage making the trip signal **DBT** go low to shut down the drive. The trip light turns on and transistor, T5 turns off to drop out the RX relay.

The TRIP potentimeter sets the TV voltage to cause a trip if a specific overcurrent condition exceeds the time rating specified for the resistor.

The equations for the differential amplifier stage and the integrator stage are listed below. Note that the amplifier signals are referenced to a fixed reference level of +7.5 volts.

**DB resistor voltage feedback (IC3[7] voltage):**

**for Vin = 600V**

VFB = 7.5 — (22.1/3039 x Vin)

= 7.5 — ($7.27 \times 10^{-3} \times 600$)

= 7.5 — 4.4 = 3.1 volts

**for Vin = 300V**

VFB = 7.5 — (22.1/1539) x Vin)

= 7.5 — ($14.36 \times 10^{-3} \times 300$)

= 7.5 — 4.4 = 3.1 volts

**Integrator voltage (test post IV):**

IV = 7.5 — (BV — 7.5 + VFB — 7.5) x t/RC

= 7.5 — (BV + VFB — 15) x t/3.3

With zero voltage across the DB resistor, Vin = 0, VFB = 7.5 volts and the bias voltage BV will drive the integrator output down to about 7 volts (approximately a diode drop below the 7.5 volt reference).

The nominal voltage ranges of the potentiometers are —

BIAS, BV: 7.5 + .3 to 7.5 + 4.6 volts
(or 7.8 to 12.1 volts)

TRIP, TV: 7.5 + .47 to 7.5 + 7.06 volts
(or 7.97 to 14.56 volts)

**NOTE**

The BIAS adjustment covers an input voltage range of (.3/4.4) x 600 = 41 volts to (4.6/4.4) x 600 = 627 volts for 600VDC or (.3/4.4) x 300 = 21 volts to (4.6/4x4) x 300 = 314 volts for 300VDC.

The integrator will not ramp positive until the VFB signal is reduced ***below*** the 7.5 reference by an amount larger than the amount BV is set above 7.5 volts.

*The expression for the BV and TV bias voltage settings are -*

BV = 7.5 + (7.27 x $10^{-3}$) x **VR** x k) for 600VDC
or
BV = 7.5 + (14.36 x $10^{-3}$) x **VR** x k) for 300VDC

TV = 7.5 — (BV + **VFBT** — 15) x T/3.3

with
**VFBT** = 7.5 — (7.27 x $10^{-3}$) x **VT** for 600VDC
or
**VFBT** = 7.5 — (14.36 x $10^{-3}$) for 300VDC

where
**VR** = rated DB resistor voltage = rated res. current x res. ohms

**VT** = resistor overvoltage for a specified time T
= overload current x res. ohms for time T

T = overload resistor time period in seconds.

**VFBT** = feedback voltage during overload condition

k = 1.0 to 1.15, continuous operating margin

Example: DB resistor - 8.5 ohms, 9.2 A continuous with a 200% (18.4A) for 15 seconds short-time rating.

Let k = 1.10 to allow a 10% overload before integration is initiated.

Calculate rated resistor voltage
**VR** = 9.2 x 8.5 = 78.2 volts

Calculate the short time resistor voltage
**VT** = 18.4 x 8.5 = 156.4 volts

Calculate the bias setting
BV = 7.5 + (7.27 x $10^{-3}$ x 78.2 x 1.1) = 8.13 volts for 600 VDC
BV = 7.5 + (14.36 x to $10^{-3}$ x 78.2 x1.1) = 8.74 volts for 300VDC

Calculate feedback voltage for 200% overload conditions

**VFBT** = + 7.5 —(7.27 x $10^{-3}$ x 156.4) = 6.36 volts for 600 VDC
**VFBT** = + 7.5 — (14.36 x $10^{-3}$ x 156.4) = 5.25 volts for 600VDC

Calculate the trip setting
TV = 7.5 — (8.13 + 6.36 —15) x (15/3.3) = 9.8 volts for 600VDC

TV = 7.5 (8.74 + 5.25 — 15) x (15/3.3) =) = 12.1 volts for 300VDC

With power applied to the module adjust the BIAS potentiometer for a voltage reading of 8.13 volts for (600 VDC, or 8.74 for 300VDC) between posts BV and COM, and adjust the TRIP potentiometer for a voltage of 9.8 for 600VDC, or 12.1 for 300VDC volts from TV to COM.

## TROUBLESHOOTING

Also refer to the Inverter Troubleshooting section of the main instruction book.

If abnormal operation is suspected, turn power off, disconnect the DB module from the Inverter as described in the Start-up section and verify that the Inverter operates the motor properly.

With power off visually inspect the DB module and the DB resistor for any obvious damage. Then with an ohmmeter check for damaged power components.

## NOTE

When an SCR is fired or is turned off abrupt changes in voltage levels (high dv/dt) will occur in the power circuitry and result in electrical "noise" or transients being conducted throughout the power circuitry as well as being transmitted (by inductive or capacitive coupling) into the control circuitry. The magnitude of such transients are reduced by use of RC snubber circuits and reactors in the power circuitry. Signal wires are usually twisted wires routed away from power wiring to minimize noise pickup. Electrical noise (transients) may cause improper firing of SCRs resulting in misoperation and possible drive shutdowns due to fault currents. Noise may also cause nuisance tripping by triggering of fault detection circuits without the actual occurance of any real faults. Also consider the possible effect of noise transients generated external to the drive. Some typical sources are operation of unsuppressed relays and contactors, switching power factor correcting capacitor, opening of transformer primary windings, arcing faults, lightening strikes and operation ofother SCR drives.

**1. Overload trip:, TOL light on the Slowdown Module turns on during dynamic braking.**

A. If the DB resistor overheats, it's watt rating may be too low for the load inertia and deceleration time required.

B. The BIAS and/or the TRIP potentiometer is improperly adjusted. Refer to the Overload Protection section.

Verify that the drive will decelerate at a slower rate by turning the DTIM potentiometer on the Inverter Regulator Card clockwise.

**2. Overcurrent trip of inverter.**

A. The DB SCR is not commutated off. The DB resistance could be too low.

B. The inverter goes into current limit. Current limit operation **must** be avoided during slowdown or stopping. Increase the deceleration time adjustment and/or increase the current limit setting.

**3. Inverter decelerates too slow.**

A. DB resistance too high or resistor open.

B. DB SCR not turning on.

C. The commutation SCR turns on prematurely, the CVR voltage level stays high preventing another firing of the DB SCR and the Inverter decelerates slower controlled by the slowdown control circuit in the driver.

**4. Inverter trips out on starting.**

A. The DB SCR is shorted or turns on by failing to block forward voltage.

**5. General**

The DB Module operation can be checked out with the motor **disconnected**. With the Inverter running, a small reduction in reference will cause the DB module to start operating in an on-off oscillatory condition around the set reference.

## CAUTION

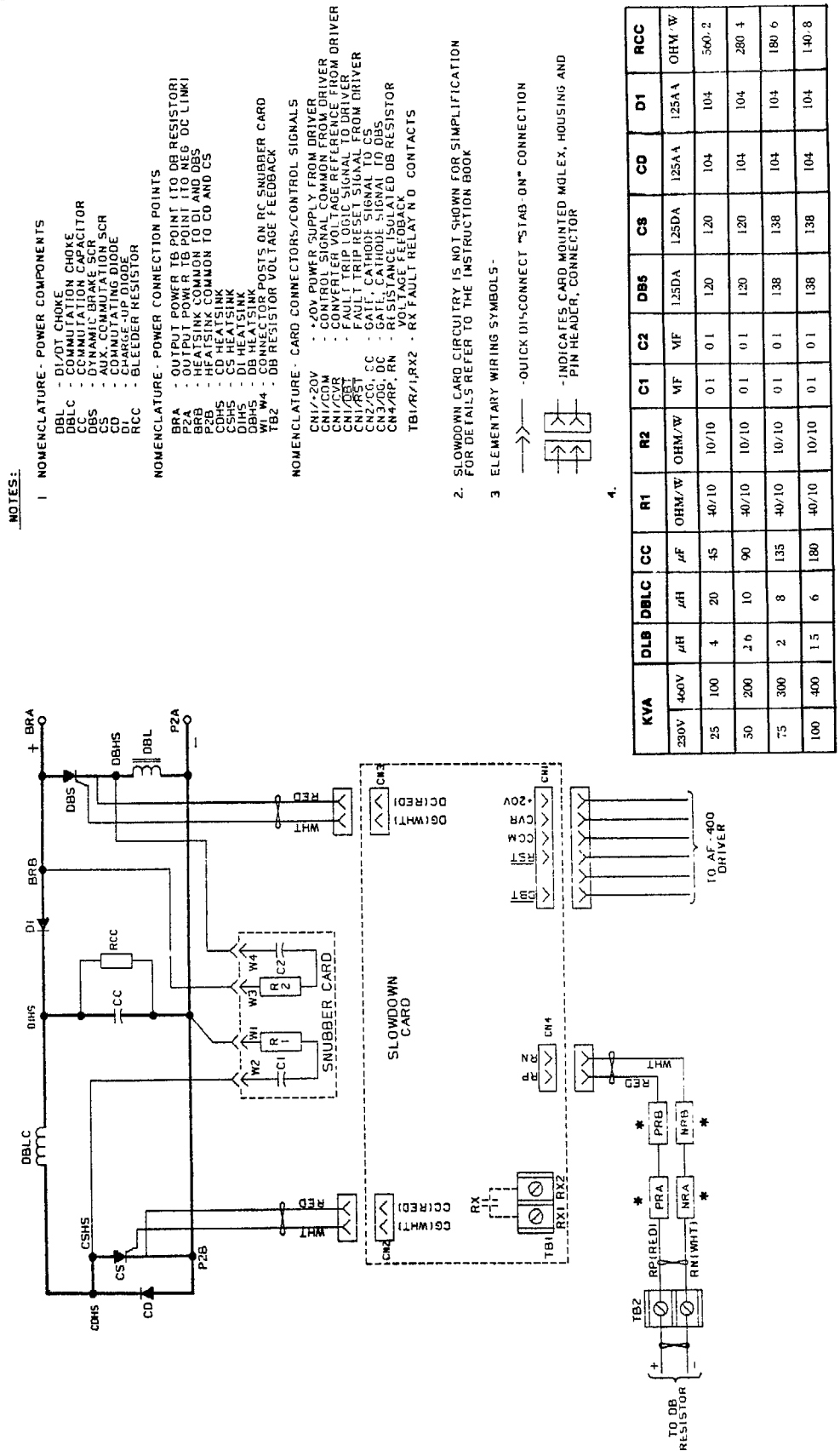
USE ONLY A DIFFERENTIAL CONNECTION (TWO PROBES WITH NO COMMON CLIPS) WHEN LOOKING AT ANY VOLTAGE LEVELS IN THE POWER CIRCUITRY WITH AN OSCILLOSCOPE.

**NOTE:** THE GATE-CATHODE CONNECTIONS CG, CC AND DG, DC ARE AT HIGH VOLTAGE WITH RESPECT TO COMMON.

The SCRs are fired by single pulses (about 20 microseconds wide) which may therefore be difficult to detect on a scope.

The safest way to check SCR firing pulses is to look at the collector voltage of transistors T2 and T4 with respect to common. (The large cooling tab of the transistors are collectors.) This voltage will notch from +20V common for about 20 microseconds during the SCR firing and then immediately followed by a reset pulse exceeding the 20V level.

To check actual gate signals a current probe should be used. (Looking at the gate-cathode voltage differently usually results in a distorted signal due to high common mode switching voltages.)




| KVA | | DLB | DBLC | CC | R1 | R2 | C1 | C2 | DB5 | CS | CD | D1 | RCC |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 230V | 460V | μH | μH | μF | OHM/W | OHM/W | MF | MF | 125DA | 125DA | 125AA | 125AA | OHM/W |
| 25 | 100 | 4 | 20 | 45 | 40/10 | 10/10 | 0 1 | 0 1 | 120 | 120 | 104 | 104 | 560 2 |
| 50 | 200 | 2 6 | 10 | 90 | 40/10 | 10/10 | 0 1 | 0 1 | 120 | 120 | 104 | 104 | 280 4 |
| 75 | 300 | 2 | 8 | 135 | 40/10 | 10/10 | 0 1 | 0 1 | 138 | 138 | 104 | 104 | 180 6 |
| 100 | 400 | 1 5 | 6 | 180 | 40/10 | 10/10 | 0 1 | 0 1 | 138 | 138 | 104 | 104 | 140 8 |

*On 460VAC Modules, resistors PRA, PRB, NRA, NRB are located in the wire harness and each 1 5 ohm

On 230VAC Modules, resistors PRB and NRB are deleted, leaving only PRA and NRA, each 1 5 OHM

**FIGURE 7**
MODULE ELEMENTARY DIAGRAM (36C766711AA)

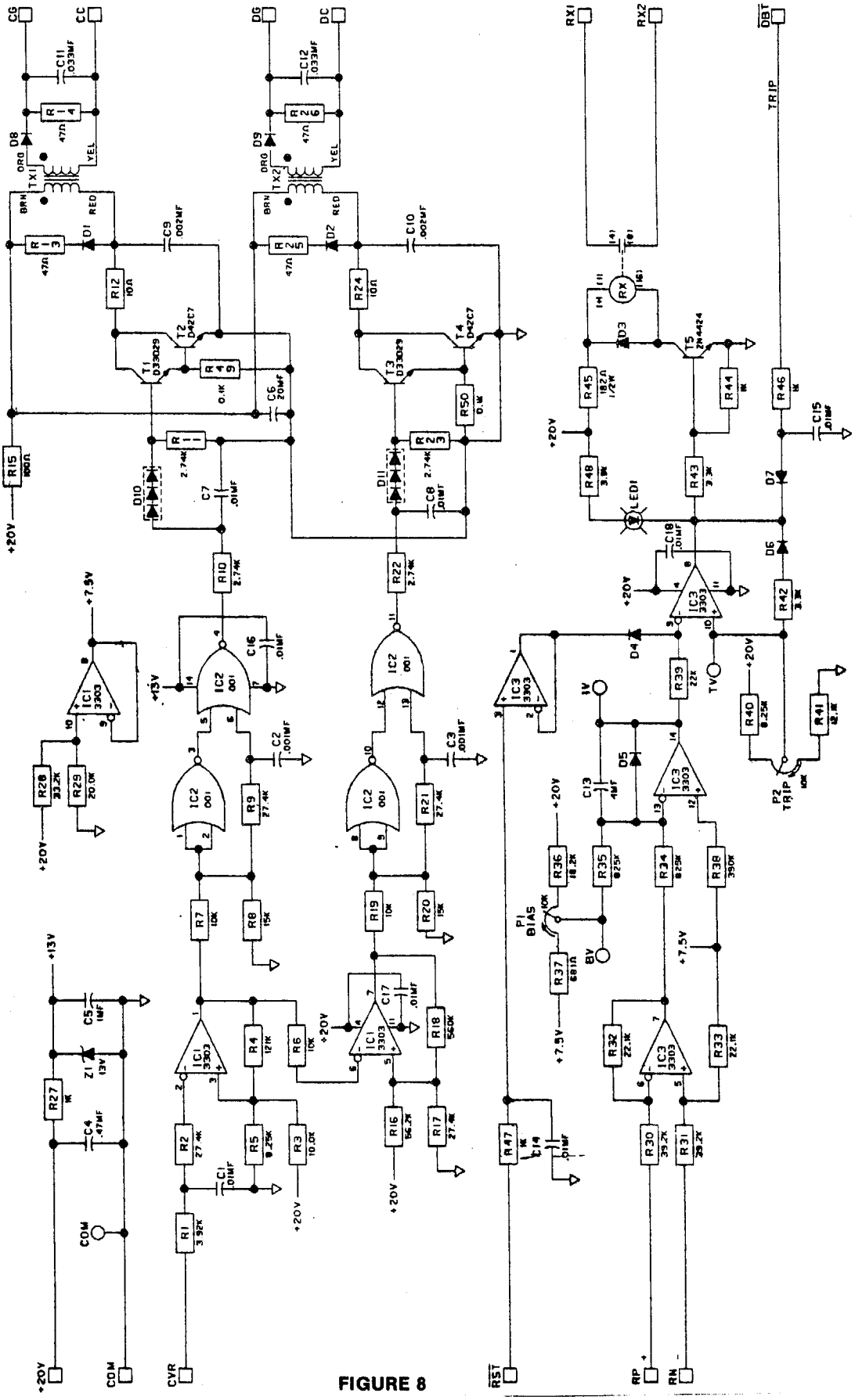


**FIGURE 8**
CARD ELEMENTARY DIAGRAM (36D871021AA)

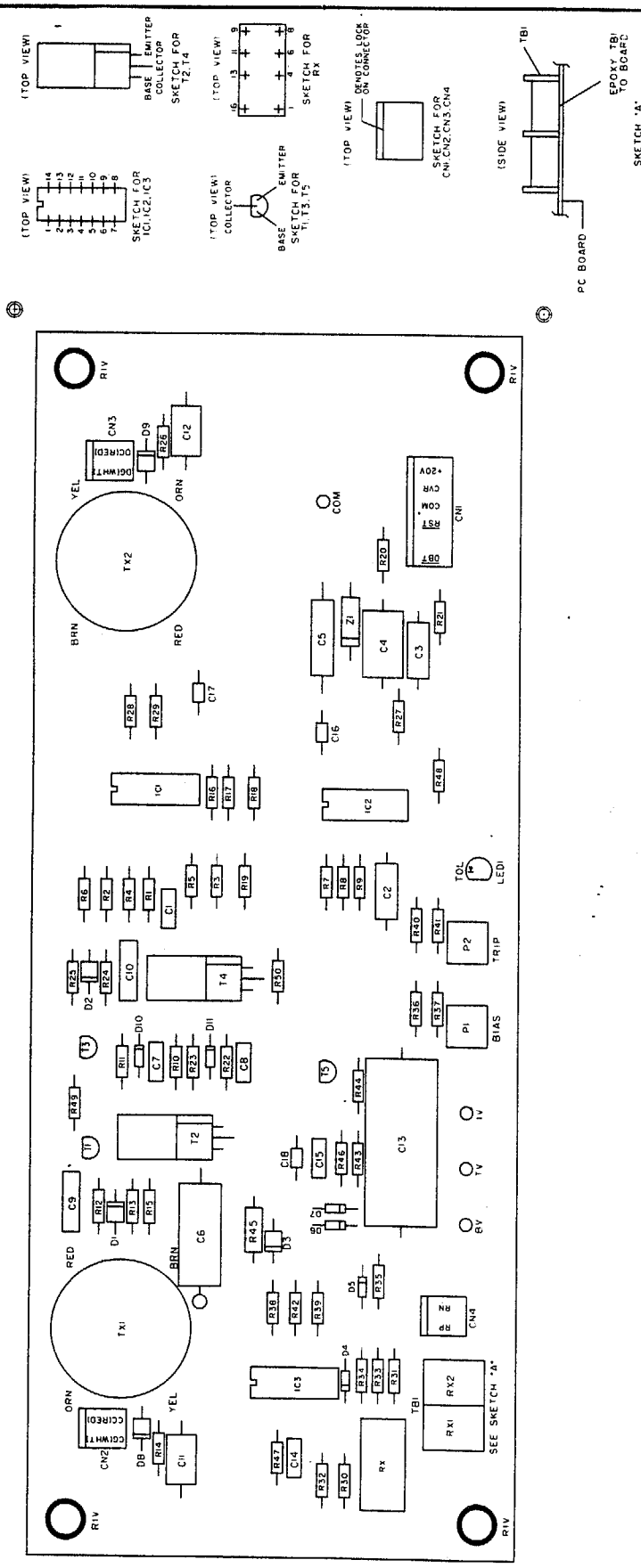



**FIGURE 9**
CARD LAYOUT DIAGRAM (36B590280AA)

GENERAL ELECTRIC COMPANY
SPEED VARIATOR PRODUCTS OPERATION
ERIE, PENNSYLVANIA 16531

GENERAL ELECTRIC

GEK-85704A (11/85) 1M (P)